



Accelerated Database Recovery – A Deep Dive Behind the Magic

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 He/Him

I have worked in the IT realm for over 27 years specializing in SQL Server for the past 16 years.

Basically, I'm a nerd

MVP – Data Platform

Friend of Redgate

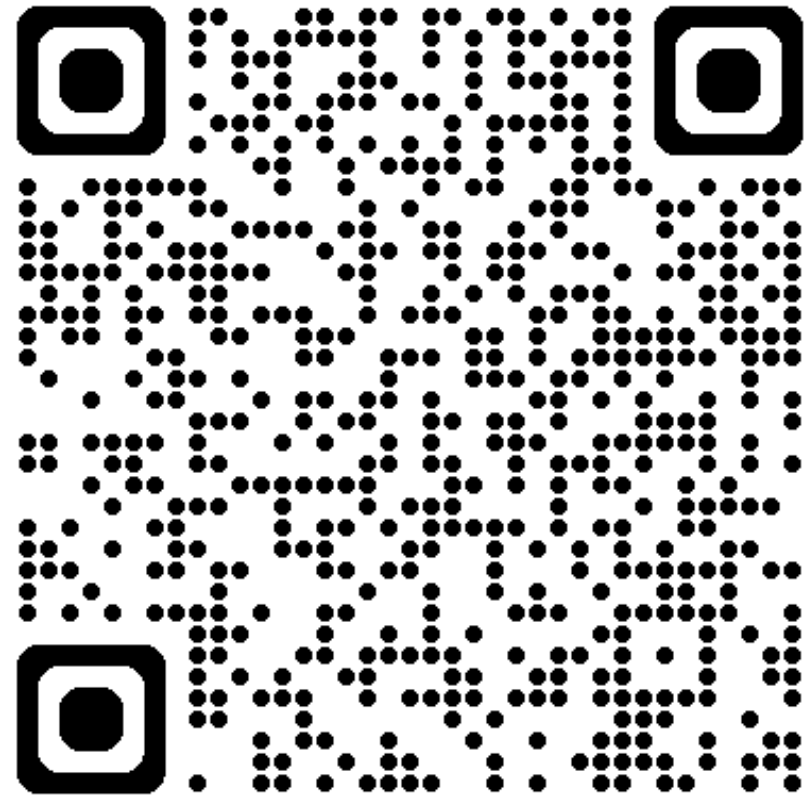
My Slides & Demos



<https://bit.ly/mypresentationfiles>



<https://github.com/eightkb/2026-August>



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ABOUT

EightKB is dedicated to offering high-quality content to the SQL Server community.



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EightKB "Melting Brain"



Premium Unisex Tee



Pullover Hoodie



Women's Slim Fit Tee

Objectives



What is it & why do we care

Components

Rollbacks/Recovery

Demo

Q&A





Constant Time Recovery in Azure SQL Database

<http://bit.ly/SQLADRExplained>



What is it?



"Accelerated database recovery (ADR) improves database availability, especially in the presence of long running transactions, by redesigning the SQL database engine recovery process."



Why do I
even care?



STILL WAITING FOR ROLLBACK TO FINISH

Accelerated Database Recovery



Faster and consistent database recovery

Almost instantaneous transaction rollback

Available in Standard Edition

Foundational piece for optimized locking

Default in Azure SQL Database/Managed Instance

Azure SQL DB



```
1  
2 DECLARE @version NVARCHAR(124)  
3 SELECT @version = @@VERSION  
4 PRINT @version;  
5 go  
6 ALTER DATABASE Demo1 SET ACCELERATED_DATABASE_RECOVERY = OFF;
```

70 %

Messages

Microsoft SQL Azure (RTM) - 12.0.2000.8
Sep 18 2021 19:01:34
Copyright (C) 2019 Microsoft Corporation

Msg 40517, Level 16, State 1, Line 6

Keyword or statement option 'accelerated_database_recovery' is not supported in this version of SQL Server.

Completion time: 2021-12-10T21:43:18.4331296-05:00



New Components

ADR New Components

Persisted Version Store

sLog

Aborted Transaction Map

Logical Revert

Cleaner



YAY FOR ADR!!!

Persisted Version Store



Same version store when using RCSI

Retains off-row versions

Lives in the user database now

Availability Group aware



In-Row vs Off-Row versions



In-row versions are kept on the page itself

In-row versions are delta changes between old/new versions

If the row change is too large, it's sent to the PVS





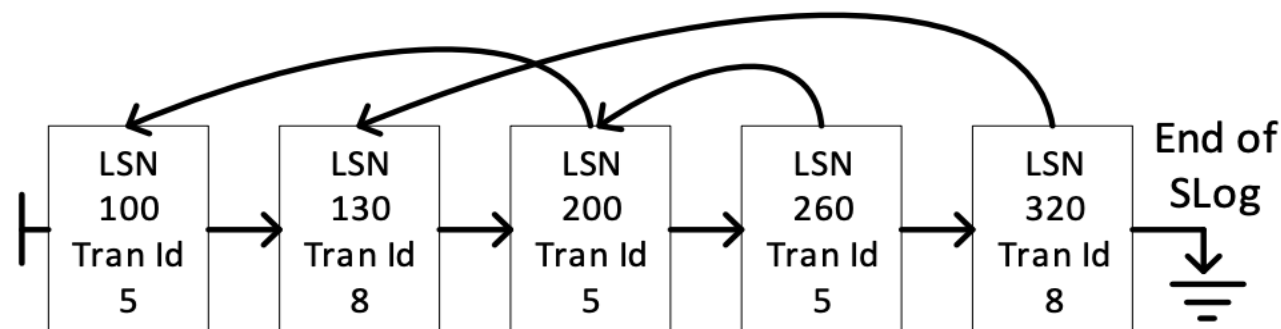
In-memory object & stores linked list of log records in order of LSN

Contained non-versioned operations

Checkpoints persist contents to disk

Provides aggressive log truncation by preserving only required log records

FN_DBSLOG(null,null)



Aborted Transaction Map

Not that ATM →

Tracks all aborted transactions

Implemented as a hash table

Rows are linked to PVS

DMV: `Sys.dm_tran_aborted_transactions`



Logical Revert



Facilitates the “undo” at the row-level

Asynchronous in nature

Uses the ATM to track what must be undone

Releases all locks immediately after abort

Cleaner

Removes versions that are not needed

Asynchronous process

Single Thread architecture

Cannot disable

Force clean up via

`sp_persistent_version_cleanup`
(synchronous)



2022 Cleaner Updates

Thread pool

Default = 1 thread

No more than # of logical cores

Start small, increased slowly



Additional SQL 2022 ADR Improvements



User transaction cleanup – cleans pages that can't be cleaned by the normal cleaner due to blocking on the ATM sweep

Decreased Memory Footprint for PVS – tracks PVS pages at the extent level to help reduce memory consumption in maintaining versioned pages

Transaction-level PVS – Allows the cleaner to clean up versions that belong to committed transactions regardless if aborted transaction are present resulting in reduced PVS growth

SQL 2025 ADR Improvements



Tempdb can be ADR enabled

Watch for excessive PVS growth due to PVS and RCSI running for other databases

PVS monitoring is the same as in user databases

ADR Configurations



Select * from sys.configurations where name like 'ADR%';

ADR cleaner retry timeout

ADR Preallocation Factor*

How to Enable ADR



Requires exclusive access to the database

Can specify specific/different filegroup

```
ALTER DATABASE [db_name]
```

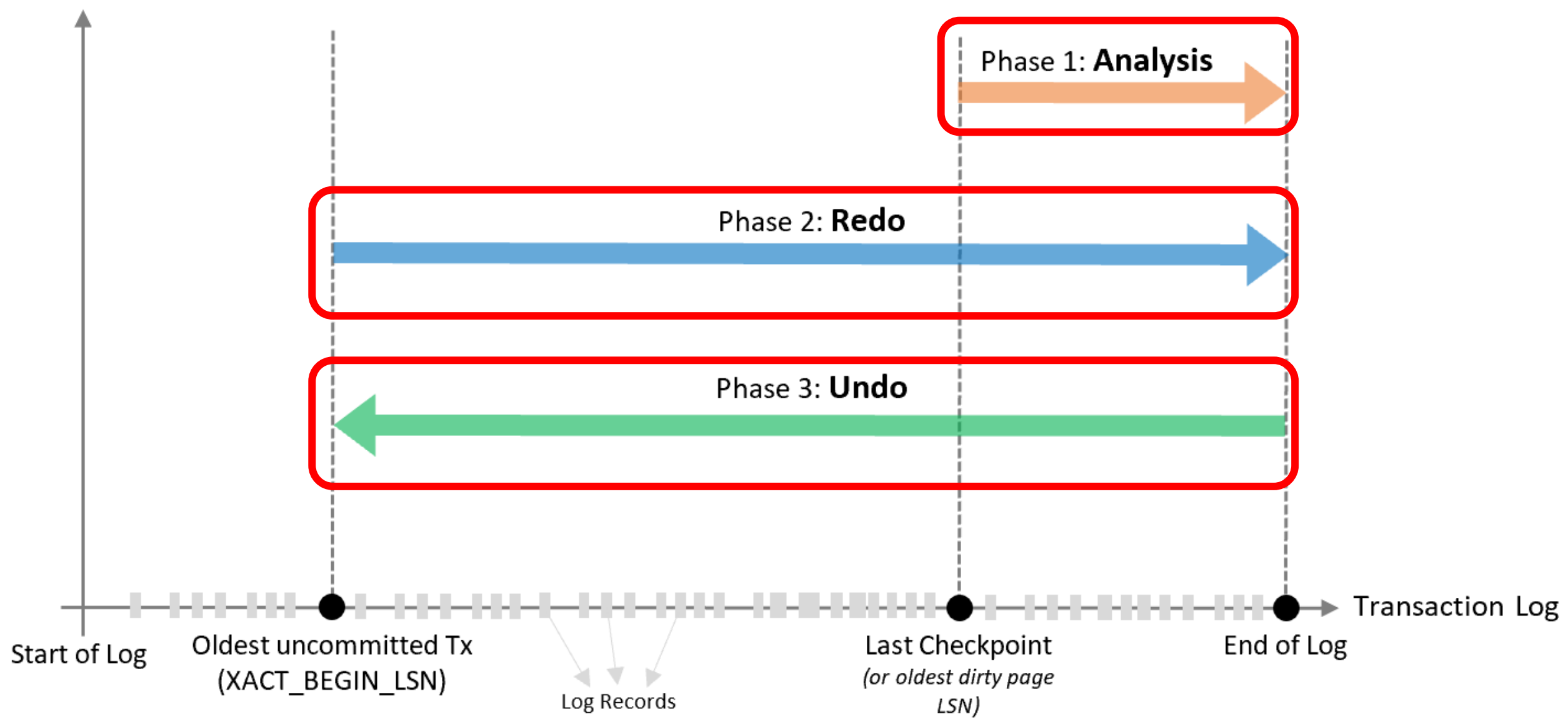
```
SET ACCELERATED_DATABASE_RECOVERY = ON
```

```
(PERSISTENT_VERSION_STORE_FILEGROUP =  
[SomeOtherFG])
```



Rollback

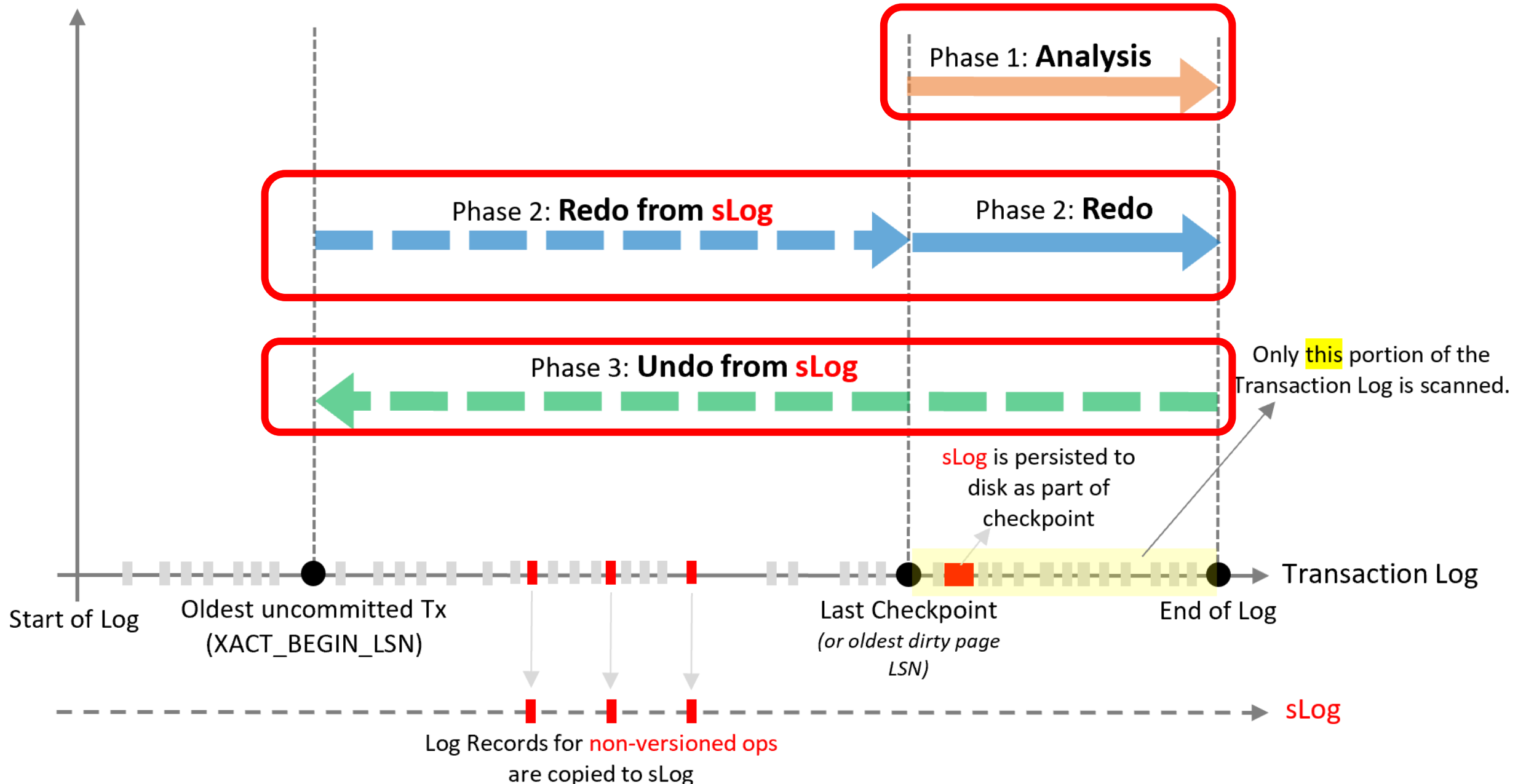
Recovery Phase / Transaction Log (without ADR)





**Rollback is
proportional to
longest running
transaction**

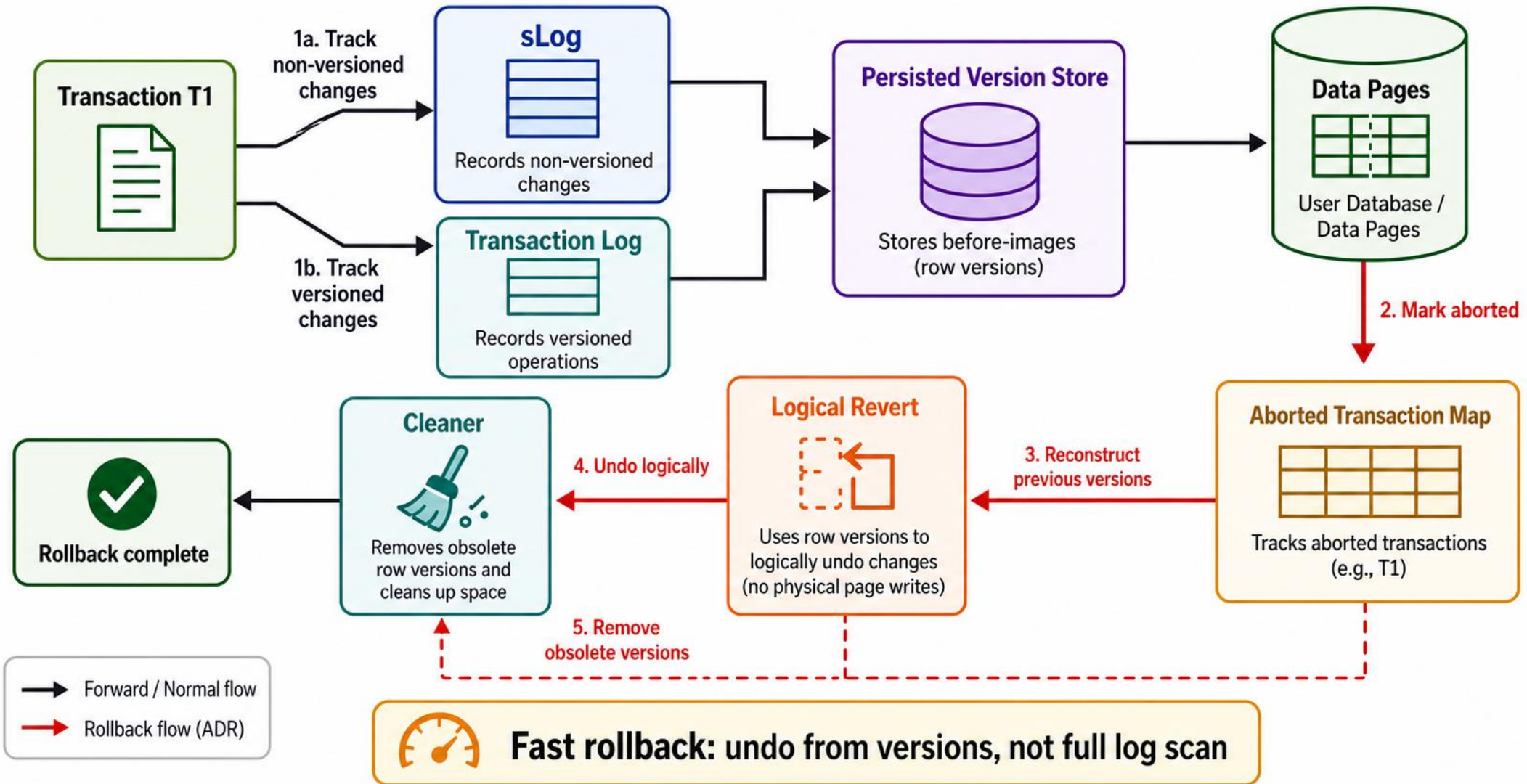
Recovery Phase / Transaction Log / sLog (with ADR)



ADR Workflow



ADR Transaction Rollback Workflow





DEMO

When Not to Deploy ADR



Extremely highly transactional

High volume of data modifications

Most likely edge case

Monitor PVS Store for issues

Transactional/Snapshot replication & CDC disable aggressive log truncation

Summary



Faster rollbacks

Faster crash recovery upon startup/failover

Aggressive log truncation

Enable for majority of use cases

Simple to enable

Resources



SQLCallStackResolver – <http://aka.ms/sqlstack>

Article - <https://www.red-gate.com/simple-talk/databases/sql-server/database-administration-sql-server/how-does-accelerated-database-recovery-work/>

Whitepaper - <http://bit.ly/SQLADRExplained>



Questions?
Answers!

**Got
Questions?**

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Slides & Demos